**Department of Mathematics and Applied Mathematics**

# Module 3MS: Metric Spaces

**Class Test One** **22 March 2016**

**Time : 18h30 – 20h00** **Total marks: 40**

1. Consider $\mathbb{R}^2$ with the usual Euclidean metric $d_E$.
   Let $A = \{(a, b) \in \mathbb{R}^2 \mid 4 \leq a < 5, b \in \mathbb{R}\}$.
   (a) Write down $\overline{A}$. (No proof required.)
   (b) Write down $\mathrm{int}(A)$. (No proof required.)
   (c) Write down $\mathrm{bd}(A)$. (No proof required.) [3]

2. Let $(X, d_X)$ and $(Y, d_Y)$ be metric spaces. For all $(a_1, a_2), (b_1, b_2) \in X \times Y$, define:

$$p((a_1, a_2), (b_1, b_2)) = \max\{d_X(a_1, b_1), d_Y(a_2, b_2)\}.$$

   (a) Prove carefully that $p$ is a metric on $X \times Y$.
   (b) Now consider the case where $(X, d_X) = (Y, d_Y) = (\mathbb{R}, d_E)$ where $d_E$ is the usual Euclidean metric on the real numbers.
   (i) Provide a sketch of the open ball, centred at the origin, with radius 1, in the space $(X \times Y, p)$. (No working required.) (ii) Provide a sketch of the closed ball, centred at the origin, with radius 1, in the space $(X \times Y, p)$. (No working required.)
   (c) Now consider the case where $(X, d_X) = (\mathbb{R}, d_E)$ where $d_E$ is the usual Euclidean metric again, but now $(Y, d_Y) = (\mathbb{R}, d_D)$ where $d_D$ is the discrete metric on the real numbers.
   (i) Provide a sketch of the open ball, centred at the origin, with radius 1, in the space $(X \times Y, p)$. (No working required.) (ii) Provide a sketch of the closed ball, centred at the origin, with radius 1, in the space $(X \times Y, p)$. (No working required.) [14]

3. Prove that, in any metric space, every open ball is an open set. [7]

4. Read the following theorem and proof, and then answer the questions that follow. Recall that, in a metric space $(X, d)$, we define the sequences $(x_n)$ and $(y_n)$ to be neighbours if $d(x_n, y_n) \to 0$ as $n \to \infty$. Recall also that for non-empty subsets $A$, $B$ of $X$, $\text{dist}(A, B) = \inf\{d(a, b) \,|\, a \in A, b \in B\}$.

**Theorem** For non-empty subsets $A$, $B$ of a metric space $(X, d)$, $\text{dist}(A, B) = 0$ if and only if there exist sequences $(a_n)$ in $A$ and $(b_n)$ in $B$ such that $(a_n)$ and $(b_n)$ are neighbours.

**Proof** Let $C = \{d(a, b) \,|\, a \in A, b \in B\}$. (1)
Then 0 is a lower bound of $C$. (2)
($\Longrightarrow$) For any natural number $n$, $\dfrac{1}{n}$ is not a lower bound of $C$. (3)
So, for any natural number $n$, there exist $a_n \in A$ and $b_n \in B$ with $d(a_n, b_n) < \frac{1}{n}$. (4)
Then $(a_n)$ and $(b_n)$ are neighbours. (5)
($\Longleftarrow$) Let $\epsilon > 0$. (6)
Choose a natural number $N$ such that $n \geq N$ implies that $d(a_n, b_n) < \epsilon$. (7)
From this, it follows that $\text{dist}(A, B) = 0$. (8)

(a) Justify line (2) by proving that $d(x, y) \geq 0$ for all $x, y$ in a metric space.

(b) Write down clearly what the statement "$\dfrac{1}{n}$ is a lower bound of $C$" means.

(c) Write down clearly what the statement "$\dfrac{1}{n}$ is not a lower bound of $C$" means.

(d) How do we know that the $N$ mentioned in line (7) exists?

(e) Explain how line (8) follows from the previous working.

(f) Now consider the case where $(X, d) = (\mathbb{R}, d_E)$ where $d_E$ is the usual Euclidean metric on the real numbers.
(i) In this space, provide examples of non-empty subsets $A$ and $B$ such that $\text{dist}(A, B) = 0$, but there do not exist elements $a \in A$ and $b \in B$ with $d(a, b) = 0$. (No proof required: just give the two sets.)
(ii) In this space, provide examples of non-empty, *closed* subsets $A$ and $B$ such that $\text{dist}(A, B) = 0$, but there do not exist elements $a \in A$ and $b \in B$ with $d(a, b) = 0$, if this is possible. If it is not possible, explain why not. [14]

5. Let $X$ be a metric space. Let $C = \{c_n \,|\, n \in \mathbb{N}\}$ be a countable subset of $X$ with $\overline{C} = X$. Find a countable collection $\{W_k \,|\, k \in \mathbb{N}\}$ of open subsets of $X$ such that, for any open subset $G$ and $x \in G$, there exists $k \in \mathbb{N}$ such that $x \in W_k \subseteq G$.
Prove that the collection $\{W_k \,|\, k \in \mathbb{N}\}$ you provide has the desired property. [2]